

Prayas JEE 2025

Maths

DPP: 3

Sequence and Series

Q1 If twice the 11^{th} term of an A.P. is equal to 7 times of its 21^{st} term, then its 25^{th} term is equal to

- (A) 24 (B) 120
(C) 0 (D) none of these

Q2 If $a_1, a_2, \dots, a_n$ are in A.P. with common difference, d , then the sum of the following series is $\sin d (\operatorname{cosec} a_1 \cdot \operatorname{cosec} a_2 + \operatorname{cosec} a_2 \cdot \operatorname{cosec} a_3 + \dots + \operatorname{cosec} a_{n-1} \cdot \operatorname{cosec} a_n)$

- (A) $\sec a_1 - \sec a_n$
(B) $\cot a_1 - \cot a_n$
(C) $\tan a_1 - \tan a_n$
(D) $\operatorname{cosec} a_1 - \operatorname{cosec} a_n$

Q3 If $a_1, a_2, a_3, \dots$ are in A.P. such that $a_1 + a_5 + a_{10} + a_{15} + a_{20} + a_{24} = 225$, then $a_1 + a_2 + a_3 + \dots + a_{23} + a_{24}$ is equal to

- (A) 909 (B) 75
(C) 750 (D) 900

Q4 Consider of A.P. with first term ' a ' and the common difference ' v '. Let S_k denote the sum of its first k terms. If $\frac{S_{kx}}{S_x}$ is independent of x , then

- (A) $a = d/2$ (B) $a = d$
(C) $a = 2d$ (D) None

Q5 The digits of positive integer having three digits are in A.P. and their sum is 15. The number obtained by reversing the digits is 594 less than the original number. Find number.

- (A) 832 (B) 752

- (C) 852 (D) 682

Q6 Let t_r denote the r^{th} term of an A.P. Also suppose $t_m = \frac{1}{n}$ and $t_n = \frac{1}{m}$, ($m \neq n$), for some positive integers m and n , then which of the following is necessary a root of the equation?

$$(l + m - 2n)x^2 + (m + n - 2l)x + (n + l - 2m) = 0$$

- (A) t_n (B) t_m
(C) t_{m+n} (D) t_{mn}

Q7 The first three terms of an arithmetic-geometric progression are 3, -1 and -1. The next term of the progression is

- (A) 2
(B) -2
(C) $-\frac{5}{27}$
(D) $-\frac{5}{9}$

Q8 The first four terms of a sequence are given by $T_1 = 0, T_2 = 1, T_3 = 1, T_4 = 2$. The general term is given by $T_n = A\alpha^{n-1} + B\beta^{n-1}$, where A, B and α, β are independent of n and A is positive. The value of $(\alpha^2 + \beta^2 + \alpha\beta)$ is

- (A) 1 (B) 2
(C) 4 (D) 5

Directions (9) Read the following passage and answer the given questions.

The first four terms of a sequence are given by $T_1 = 0, T_2 = 1, T_3 = 1, T_4 = 2$. The general term is given by $T_n = A\alpha^{n-1} + B\beta^{n-1}$, where



A, B and α, β are independent of n and A is positive.

Q9 The value of $5(A^2 + B^2)$ is

- (A) 2 (B) 3
(C) 4 (D) 5

Q10 Comprehension Type Questions

The first four terms of a sequence are given by $T_1 = 0, T_2 = 1, T_3 = 1, T_4 = 2$. The general term is given by $T_n = A\alpha^{n-1} + B\beta^{n-1}$, where A, B and α, β are independent of n and A is positive. The quadratic equation, whose roots are α and β , is given by

- (A) $x^2 - 2x - 1 = 0$
(B) $x^2 - 2x - 2 = 0$
(C) $x^2 - x - 1 = 0$
(D) None of these

Q11 If a_m denotes the m^{th} term of an AP, then a_m is equal to

- (A) $\frac{2}{a_{m+k} + a_{m-k}}$ (B) $\frac{a_{m+k} - a_{m-k}}{2}$
(C) $\frac{a_{m+k} + a_{m-k}}{2}$ (D) None of these

Q12 If the 6^{th} term of an AP is 2, then the value of the common difference of an AP, which makes the product $a_1 a_4 a_5$ least, is given by

- (A) $\frac{8}{5}$
(B) $\frac{5}{4}$
(C) $\frac{2}{3}$
(D) None of these



Answer Key

Q1 (C)
Q2 (B)
Q3 (D)
Q4 (A)
Q5 (C)
Q6 (D)

Q7 (D)
Q8 (B)
Q9 (A)
Q10 (C)
Q11 (C)
Q12 (C)



[Android App](#) | [iOS App](#) | [PW Website](#)

